# Wk8.1 SVM

Thursday, November 5, 2026 8:26 AM

Close to LR math-wise

Ch9 Support $\overset{LA}{\text{Vector}}$ Machines (SVM) → Only on classification, though SV regressors exist.

History: 1960's stat approach
1990's CS & top ML approach until early 2000s

↳ Solid theory & flexible

DTs took over

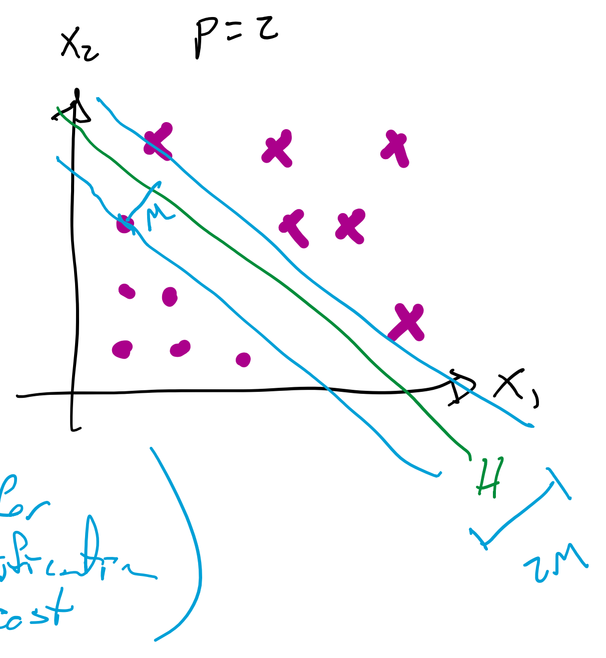
Build SVMs via 3 steps:

V1 Max-margin classifier (issue, does not allow train error)

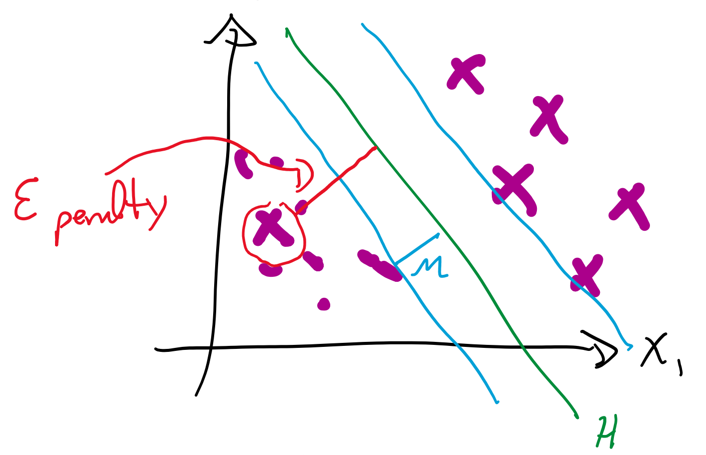
↳ Hyperplane separation (H)

Vector form.

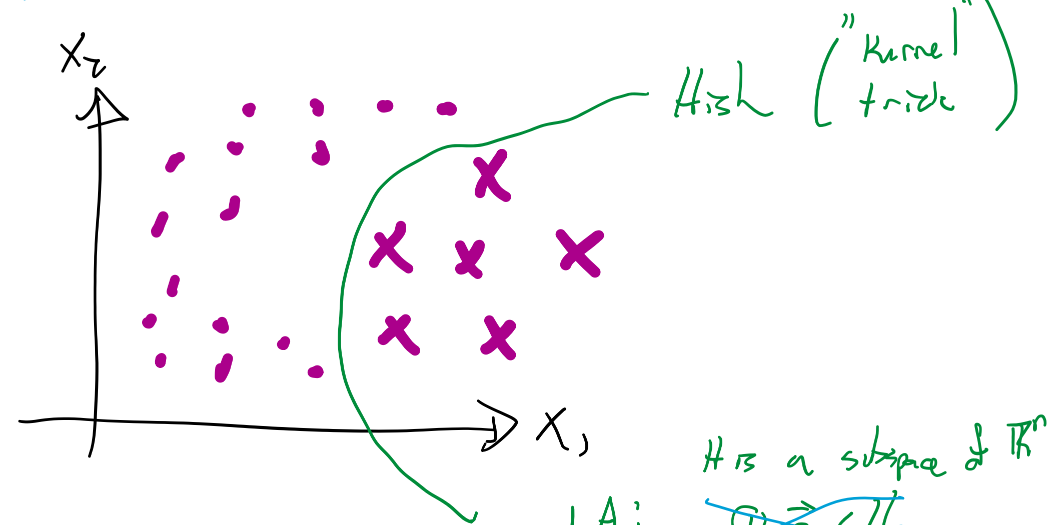
Best choice by maximize margin (M) via optimization



V2 Support vector classifier (allow for misclassification at cost)



V3 SVM (allow for non-linear separation)



LA:

H is a subspace of $\mathbb{R}^n$ if
~~① $\vec{0} \in H$~~
② If $\vec{u} \in H$, $(c\vec{u}) \in H$
③ If $\vec{u}, \vec{v} \in H$, $(\vec{u}+\vec{v}) \in H$

Hyperplane introduction (Calc 3)

In $\mathbb{R}^p$ is a (p−1) dimension "subspace" affine space

# predictors

$\mathbb{R}^2$: General form: $\boxed{\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0}$ linear regression-ish

Need point on H & vector normal to H

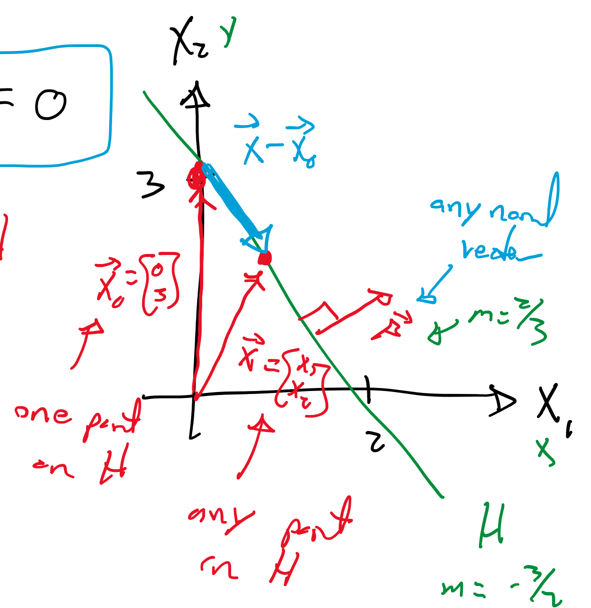
$$\vec{\beta} \cdot (\vec{x} - \vec{x}_0) = 0$$

$$\begin{bmatrix}3\\2\end{bmatrix} \cdot \left(\begin{bmatrix}x_1\\x_2\end{bmatrix} - \begin{bmatrix}0\\3\end{bmatrix}\right) = 0$$

$$\begin{bmatrix}3\\2\end{bmatrix} \cdot \begin{bmatrix}x_1\\x_2\end{bmatrix} - \begin{bmatrix}3\\2\end{bmatrix} \cdot \begin{bmatrix}0\\3\end{bmatrix} = 0$$

$$3x_1 + 2x_2 - 6 = 0$$

$$\boxed{-6 + 3x_1 + 2x_2 = 0}$$



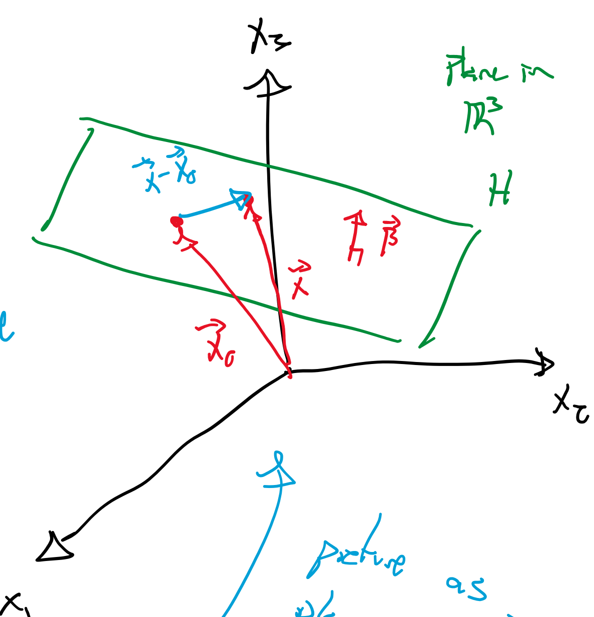
~~Ans: y = mx + b~~
~~$\boxed{y = -\frac{3}{2}x + 3}$~~ $\mathbb{R}^3$
~~2y = −3x + 6~~
~~$\boxed{-6 + 3x + 2y = 0}$~~

$\mathbb{R}^3$ $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 = 0$

$$\vec{\beta} \cdot (\vec{x} - \vec{x}_0) = 0$$

$$\begin{bmatrix}\beta_1\\\beta_2\\\beta_3\end{bmatrix} \cdot \left(\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} - \begin{bmatrix}a\\b\\c\end{bmatrix}\right) = 0$$

same



picture as plane in $\mathbb{R}^3$

In $\mathbb{R}^p$: $\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$

$\beta_0 + \vec{\beta} \cdot \vec{x} = 0$, $\vec{x} = \begin{bmatrix}x_1\\\vdots\\x_p\end{bmatrix}$, $\vec{\beta} = \begin{bmatrix}\beta_1\\\vdots\\\beta_p\end{bmatrix}$

dot prod

$\beta_0 + \vec{\beta}^T \vec{x} = 0$ same!

$\beta_0 + \langle \vec{\beta}, \vec{x} \rangle_2 = 0$ $l^2$-inner product

How to build a classifier via a hyperplane? p=2, c=2 (classes)

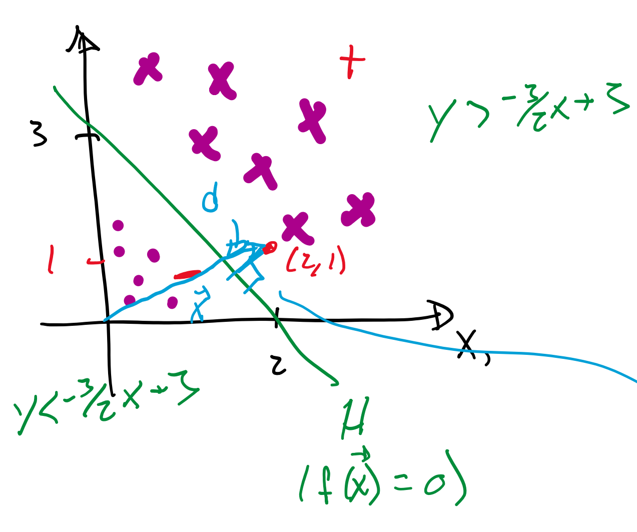
Denote $f(\vec{x}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
$= -6 + 3x_1 + 2x_2$

H: $f(\vec{x}) = 0$

H features: ① Separates $\mathbb{R}^2$ & changes sign.

2 classes: ① $f(\vec{x}) > 0$ ✖
② $f(\vec{x}) < 0$ ●

↳ No clear direction (up/down)

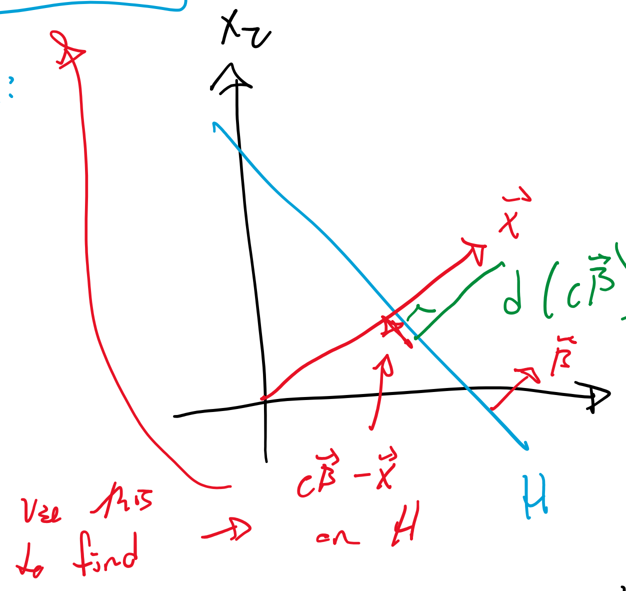


$f\left(\begin{bmatrix}2\\1\end{bmatrix}\right) = -6 + 3(2) + 2(1)$
$= -6 + 6 + 2$
$= 2 > 0$

② Use H & distance to measure classification certainty.

(orthogonal distance) $d = \dfrac{f(\vec{x})}{\|\vec{\beta}\|} = \dfrac{\beta_0 + \vec{\beta} \cdot \vec{x}}{\|\vec{\beta}\|} = \dfrac{2}{\left\|\begin{bmatrix}3\\2\end{bmatrix}\right\|} = \dfrac{2}{\sqrt{3^2+2^2}} = 2/\sqrt{13}$

Idea:



H is found by maximizing the smallest d & separating classes.

V1 Max margin classifier

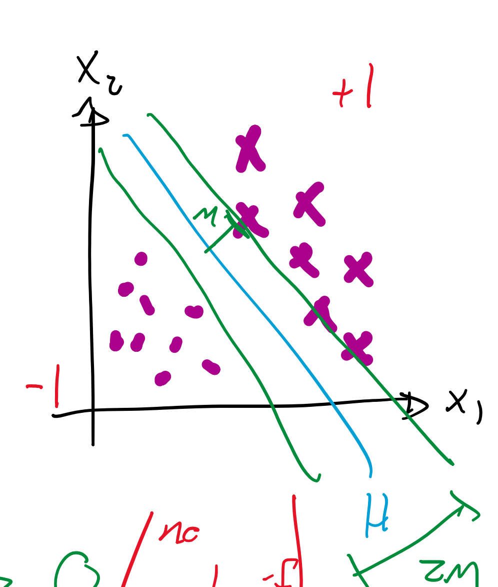
$\max_{\beta_0, \vec{\beta}} M$, $M_i = \dfrac{f(\vec{x}_i)}{\|\vec{\beta}\|} = \dfrac{\beta_0 + \vec{\beta} \cdot \vec{x}}{\|\vec{\beta}\|}$ train samples $\vec{x}_i$

$M = \min_i M_i$

subject to $y_i f(\vec{x}) = \overset{\pm}{y_i}(\beta_0 + \overset{\pm}{\vec{\beta}}\,\vec{x}) > 0$ (no misclassif)

+1 or −1

Modify to allow misclassify (SVC) next time!



Compute via constrained optimization (numerical approx & mostly exact approaches) Skip.

Also chose $\|\vec{\beta}\| = 1$ to simplify some.